

Can a neural Willmore surface be trusted?

Certificates and adversarial tests from the analysis of the Willmore functional

Nina da Hora, MSc, Institute of Computing (IC), Unicamp
MM845 AI for Geometry, mini-project. IMECC, Unicamp, October 2026
Lecturers: Henrique N. Sá Earp, Edward Hirst, Tomás S. R. Silva

The neural Willmore flow: method and unchecked properties

Hirst, Sá Earp, Silva. Minimising Willmore energy via neural flow, arXiv:2604.04321 (2026)

What the method does	What is never checked on its output
A tanh network $\phi(u, v) \rightarrow \mathbb{R}^3$ on spectral features; curvatures by automatic differentiation.	The topology (is the output still a torus?).
Loss: Monte Carlo estimate of $\int H^2 dA$ on 5000 uniform random points, with a Huber cap on H^2 at 50 and regularity terms; Adam.	The Willmore equation (is it a critical point?).
Recovers 4π (sphere) and $2\pi^2$ (Clifford torus); reports $W = 30.2$ for genus 2.	Mobius invariance of the reported energy; embeddedness (no self-intersection term).
One run per genus, one seed, no error bar.	The estimator itself: is the reported number the energy of the surface?

The question of this project: can the analysis of the Willmore functional, made computable, certify a numerical surface, and where does the training signal fail?



Certificates derived from the analysis of W

An identity with an exact reference value, computed on a grid the training never saw. A residual size is a diagnostic, reported with a band, not pass or fail.

No.	Theorem	Computed on the network output	Pass rule
1	Gauss-Bonnet	$\int K \, dA$ on the frozen grid	$ \int K \, dA - 4\pi(1 - g) < 1e-3 \times 2\pi$ on a converged grid
2	Mobius invariance	W after two inversions, centres outside and at the centroid	relative difference $< 1e-3$
3	Li-Yau	certified W against 8π ; triangle mesh of the output tested for self-intersection	$W < 8\pi$ certifies embeddedness; a crossing with $W < 8\pi$ convicts the estimator; above 8π only the mesh speaks
4	Willmore equation	$L2$ norm of $\Delta H + 2H(H^2 - K)$: direct (4th derivatives) and variational (2nd derivatives)	relative: < 0.05 critical, < 0.5 near-critical, else not critical
5	Energy quantisation	the attack: a grafted bubble $\phi + \epsilon \exp(-d^2/\rho^2) n$; reward = $W_{\text{cert}} - W_{\text{Huber}}$	an accepted surface with reward $> 2\pi$

Course material used: Lectures 4, 12, 13 (the neural flow, PINNs, geometric flows), 2 (baselines, the flow's metric, optimisers), 3 (linear ansatz), 9 (search ladder as adversary), 7 and 8 (basin map). Lectures 5, 6, 10, 11 have interfaces written, not run.

Reported energy and certified energy

	Reported energy, W_{rep} (the training signal)	Certified energy, W_{cert} (the harness)
How	domain area x mean of $H^2 \sqrt{g}$ over 5000 uniform random points; Huber cap on H^2 above 50; one seed	int H^2 dA by spectral quadrature in float64 (midpoint rule in periodic directions, Gauss-Legendre toward the poles), grid refined until Gauss-Bonnet closes to $1e-6$
Weakness	random samples miss small features; the cap hides large curvature; the value moves with the seed	needs a grid that resolves the surface; the resolution protocol guarantees it, with Gauss-Bonnet as the criterion
Check	none internal	an exact zero: Gauss-Bonnet on the same grid, $1e-15$ on the checkpoints

- Harness: certificates.py, any differentiable $\phi(u, v) \rightarrow \mathbb{R}^3$ in torch; same autodiff curvatures as the paper; nothing clamped (a NaN is a finding).
- Validated twice before touching a network: numpy with finite differences and torch with autodiff agree to $1e-10$ on the exact sphere, ellipsoid, Clifford torus and a torus of the wrong aspect ratio (18 tests, 4 s). The wrong torus passes Gauss-Bonnet and fails the Willmore equation: right topology is not Willmore (appendix).
- The gap $W_{\text{cert}} - W_{\text{rep}}$ is not a certificate. It measures how far the training signal can be pushed from the truth; the attack maximises it.

Data and runs

Everything shown was run.

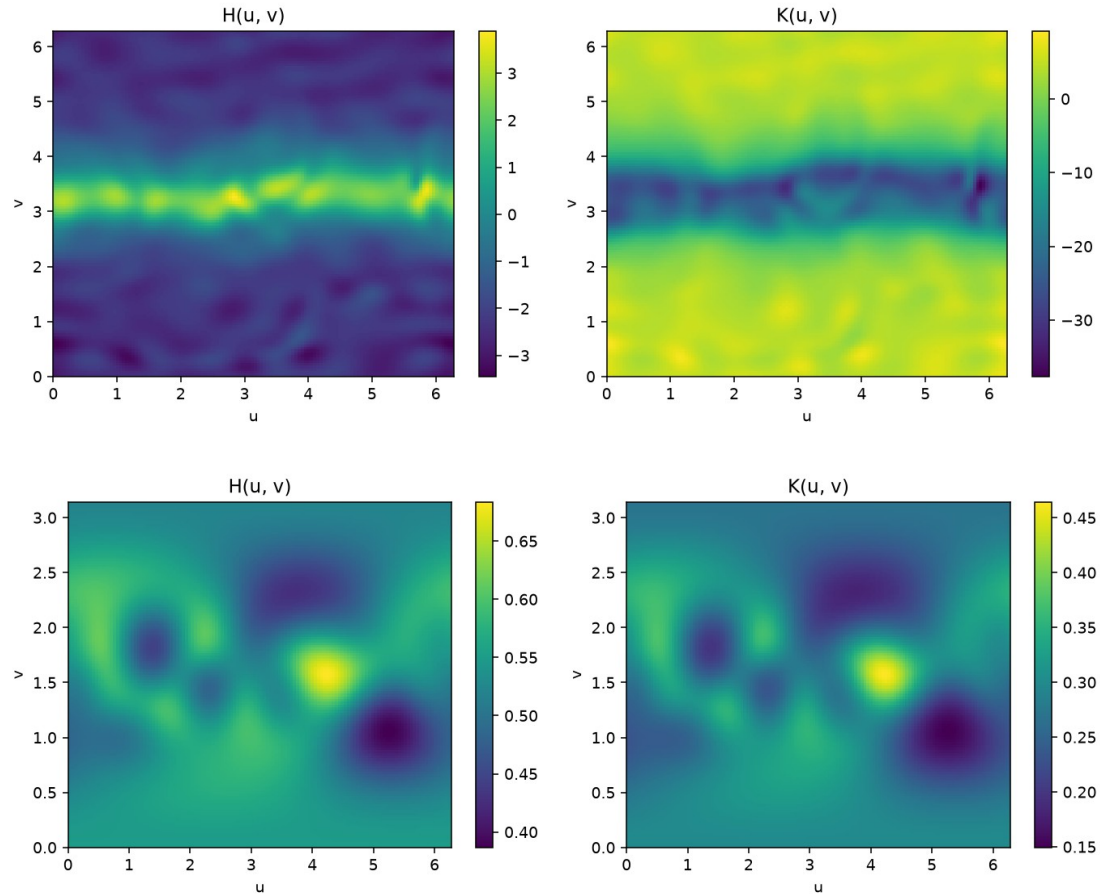
What	How many	Role
Exact surfaces: sphere, ellipsoid (1,1,2), Clifford torus, torus $R = 3$	4	unit tests of the harness; the trivial baseline (Lecture 2)
Outputs of the published flow, trained with the released code and configs	genus 0 (30 epochs), genus 1 (200 epochs, best at 141)	the objects audited
Grafted bubbles on the trained torus	16 (4 heights x 4 widths), 5 seeds each	the attack (certificate 5)
Training-estimator replicas	5 seeds x $n = 5000$ per surface	bias, spread, Huber effect
Alternative methods under the same audit	linear spectral ansatz; published flow with 2 more seeds (the residual-loss PINN did not run)	the ingredient table
Perturbed initial tori through the published flow	6 Fourier-perturbed, 2 twisted tau, 1 (2,3) torus knot	a first basin map

Evaluation grids: 48^2 to 384^2 tensor grids, never used in training. Genus 2 (the June reproduction, $W_{rep} = 29.70$ and 29.23) is not audited yet: the two-chart quadrature is not written.

Tooling. The harness code was drafted with an AI assistant (Claude) from my specification; the protocol text and slides were organised with its help. The certificates, thresholds, experimental decisions, all runs and the interpretation are mine. Numerical claims were cross-checked by two independent implementations (numpy and torch) agreeing to $1e-10$ on exact surfaces; 18 unit tests pass.

Certificate 4: Willmore equation residual

$r = \Delta H + 2H(H^2 - K)$; bands: critical < 0.05 < near-critical < 0.5 < not critical



H and K of the trained torus (top) and sphere (bottom), on the parameter grid

Checkpoint	W_cert / min	Direct	Variational
Torus, run_1	1.0117	0.99, not critical	0.21, near-critical
Sphere, run_2	1.0025	0.94, not critical	0.22, near-critical

- Torus: 98.1% of H's spectral energy is below mode 6 (the network's Fourier range); the 1.9% above is the mottling in the map, amplitude 0.05. A ripple of amplitude a and frequency k costs a^2 in energy and a k^4 in ΔH : the energy is blind to it, the equation is not.
- Sphere: smooth by construction (harmonics up to degree 2), simply not round. Near a minimum the energy is quadratic in the deviation (0.25% in W), the residual is linear (20%). The energy criterion is second-order blind where the equation sees first-order.
- Variational measurement (second derivatives only): near-critical in the smooth directions, not critical in the fine ones.

The training estimator: spread, bias and epoch selection

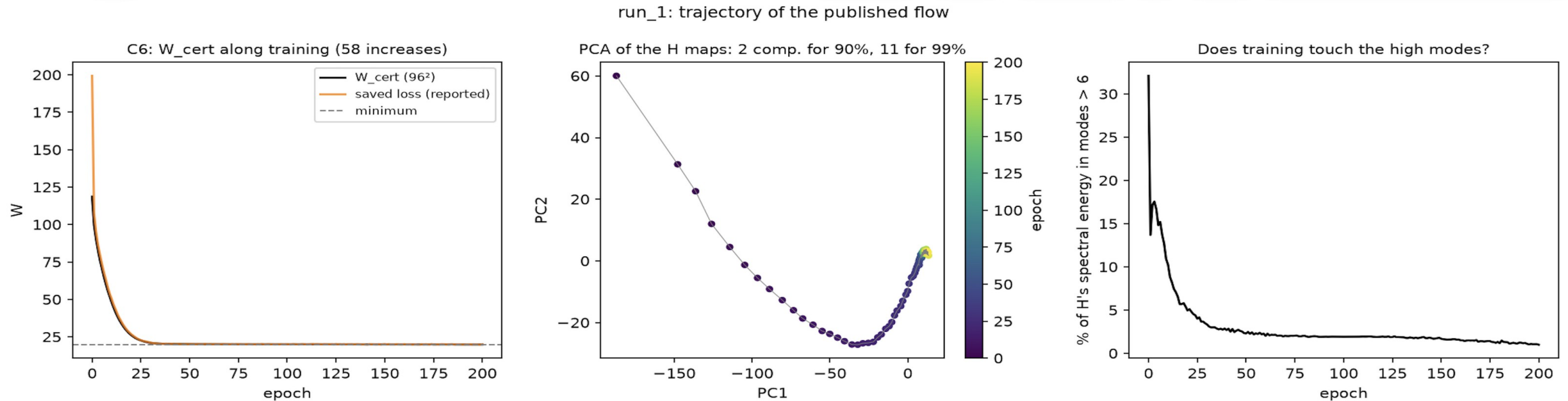
5 seeds at $n = 5000$, the published setting. Bias is significant above $2 \text{ sd} / \sqrt{5}$. Two surfaces within 2 sd are a tie.

Surface (published flow, unchanged)	W_cert (x min)	W_rep, 5 seeds	Reported by the training	Reading
Torus, run_1, best-by-reported epoch (141 of 200)	19.9710 (1.0117)	19.906 +- 0.092	19.721	1.3% below certified; epoch 199 has W_cert 19.873 (1.0068): the selected surface is 0.5% worse
Sphere, run_2, last epoch	12.5981 (1.0025)	12.633 +- 0.068	12.735	1.1% above certified
Sphere, run_2, best epoch by reported energy	12.6997 (1.0106)	12.749 +- 0.090	12.424 = $0.9887 \times 4\pi$	reported below the minimum; the selected surface is worse than the last epoch
Torus from a Clifford-shaped initial surface (6 runs)	1.0014 to 1.0021	spread 0.07 to 0.17	19.51 to 19.57, all below $2\pi^2$	every best epoch is a draw below the proven minimum (Marques-Neves)
Torus, two more training seeds from the config's initial torus	1.0070, 1.0084	0.15, 0.15	19.657 (below $2\pi^2$), 19.812	seed spread of W_cert: 0.25%

- The reported number is a draw around the certified one with a spread of about half a percent, in both directions. Selecting the best epoch selects the lucky draw: it reports values below proven minima, and on both the sphere and the torus it kept a worse surface than the last epochs.
- Initial data matters more than the seed: from the config's thin torus the flow ends 0.7 to 1.2% above the minimum; from a Clifford-shaped start, 0.15 to 0.2%. The seed spread of the certified energy is 0.25%.
- Consequence for genus 2: the train-sample energy oscillating between 28.9 and 30.6 on consecutive epochs is this noise; 30.19 (paper) and 29.70 (June reproduction) are a tie.

Certificate 6 and the trajectory of one training run

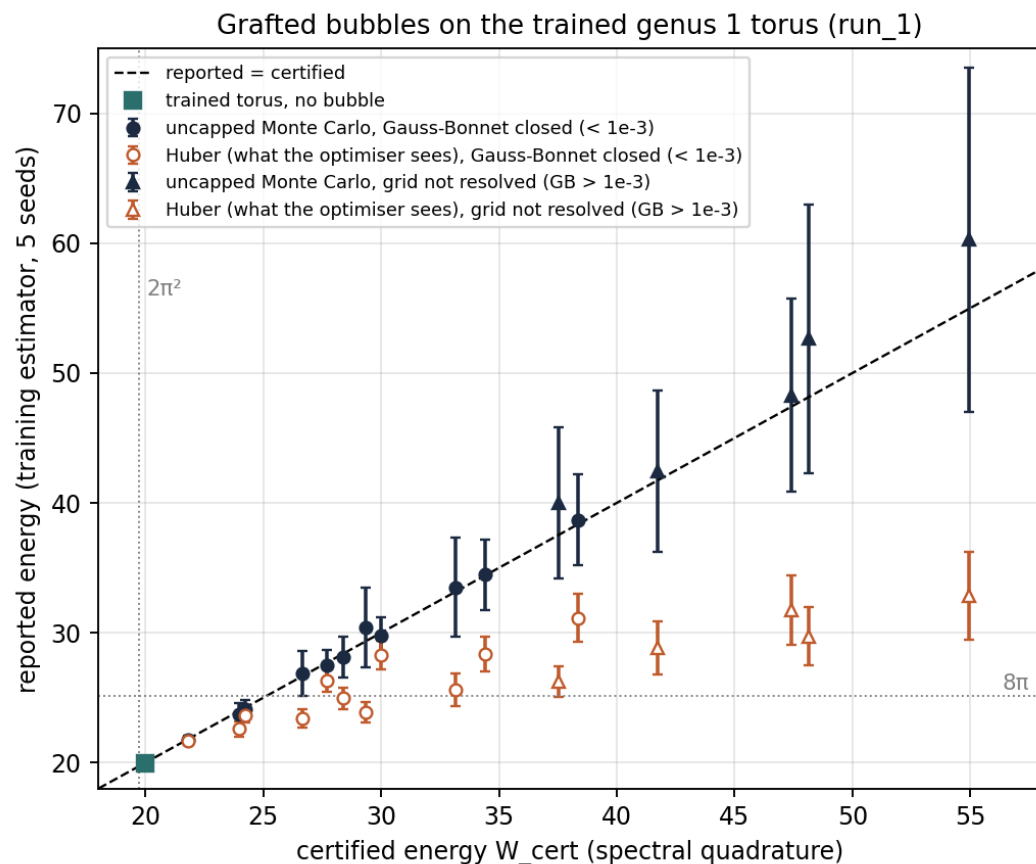
run_1, 201 checkpoints (one per epoch), W_{cert} on 96×96 and the H map on 48×48 at each; PCA of the H maps (Lectures 7, 8)



- C6 (Kuwert-Schatzle): the true Willmore flow is monotone; the certified energy of the training rises in 58 of 200 epochs. The optimiser descends on the estimator, and a quarter of its steps raise the energy. The best-by-reported checkpoint (epoch 141) is 0.5% worse than epoch 199.
- The trajectory is two-dimensional: 2 principal components of the H maps carry 90% of the motion, 11 carry 99%, in a 33 859-parameter network. Training moves on a low-dimensional manifold.
- High modes (above the feature band, $k > 6$): 32% of H's spectral energy at epoch 0, 2% by epoch 40, 1% at epoch 200. The mottling that fails certificate 4 is what the energy gradient removes last: a ripple of amplitude a^2 in W , so the flow barely sees it.

Adversarial test: grafted bubbles on the trained torus

16 bubbles on the genus 1 checkpoint ($W_{\text{cert}} = 19.971$), 5 seeds each, grid refined per bubble



eps	rho	W_{cert}	GB residual	Uncapped MC	Huber (optimiser)	Hidden
0.05	0.5	21.81	4e-15	21.75 +- 0.34	21.66 +- 0.30	0.4%
0.05	0.3	23.95	5e-15	23.68 +- 0.88	22.59 +- 0.57	4.6%
0.05	0.15	29.32	8e-6	30.42 +- 3.06	23.86 +- 0.78	21.6%
0.10	0.2	33.14	5e-6	33.50 +- 3.81	25.59 +- 1.28	23.6%
0.30	0.3	38.35	7e-7	38.70 +- 3.51	31.14 +- 1.85	19.5%

- The uncapped estimator is unbiased, but its seed spread grows from 0.3 to 13 as the bubble sharpens.
- The Huber value is what the optimiser descends on: it hides up to a quarter of the energy and saturates near 25 to 33 whatever the true value.
- Row 3: certified 29.3, above 8π ; reported to the optimiser 23.9, below 8π . A bubble of height 0.05 moves the training signal across the Li-Yau line.
- Gauss-Bonnet closes to $1e-6$ on these rows: topology certified. Mesh test: the 29.3 bubble does not self-intersect (above 8π , only the mesh can say so); the 0.3-high bubble crosses the opposite wall, and the training's regularity thresholds accept it.

Audit table and comparison criterion

One row per family of surfaces; value and verdict in every cell. 17 audited surfaces, 8 mesh-tested; full table in the repository.

Surface	Runs	W_cert (x min)	C1 Gauss-Bonnet	C2 Mobius	C3 Li-Yau	C4 residual
Published flow, sphere (last epoch / best-by-reported)	2	1.0025 / 1.0106	pass, 1e-14	1e-16	embedded; mesh: no crossing	0.94 / 0.99, not critical
Published flow, torus, config's initial torus (3 seeds)	3	1.0070 to 1.0117	pass, 4e-15	1e-16 to 2e-8	embedded; mesh: no crossing	0.95 to 0.99, not critical
Published flow, torus, Clifford-shaped initial surfaces (6 Fourier, 2 twisted)	8	1.0014 to 1.0176	pass, 5e-15	5e-15 to 3e-9	embedded; mesh: no crossing (2 tested)	0.75 to 0.95, not critical
Published flow, torus from a (2,3) torus knot tube	1	1.8818 (above 8pi)	pass, 7e-15	5e-11	W >= 8pi, theorem silent; mesh: crosses (784 pairs)	0.99, not critical
Linear spectral ansatz, no hidden layer (Lecture 3)	1	1.0033	pass, 4e-15	7e-14	embedded	0.80, not critical
Residual-loss PINN, weak form (Lectures 12, 13)	1	1.0620	pass, 4e-15	2e-16	embedded	0.98, not critical

- Every surface produced is a genuine closed surface of the right genus, Mobius-consistent; none is a critical point of W. 8 of 9 initial data reach the Clifford basin. The ninth started as an embedded knotted tube and the flow unknotted it by passing through itself: an immersed torus at 1.88 x, accepted by the training (no self-intersection term, thresholds pass), seen only by the mesh. Depth is worth about 0.15% on genus 1; the initial surface, ten times that. The residual-loss arm zeroes the first variation where it looks and still fails the direct residual: 6% above the minimum.
- Better, within a declared boundary (genus, initial data, budget): at least as good on every axis and strictly better on one. Otherwise a trade-off, and the table says which. In geometry a result holds under its hypotheses; here they are the first column.

Summary

A finer grid gives a better number; a certificate gives a reason to believe it, and in one case (Li-Yau) it can show that the estimator is wrong.

Found	Not claimed
Gauss-Bonnet fails before W converges: it is the resolution criterion. All 17 audited outputs are closed surfaces of the right genus and Mobius-consistent; none is a critical point of W . The reported energy is a draw around the certified one (half a percent); best-epoch selection picks the lucky draw, reports values below proven minima, and on the sphere kept a worse surface. The certified energy rises in a quarter of the training steps (C6) and the trajectory is two-dimensional. Initial data matters ten times more than the seed. From a knotted start the flow crosses itself and stalls immersed at 1.88 x; nothing in the training notices, the mesh does. A bubble of height 0.05 hides 22% of the energy from the optimiser.	Anything about the genus 2 minimiser; optimality of any method in general; that a passing certificate proves the surface is the minimiser. A pass means: consistent with a Willmore surface of that genus at that resolution. Next steps and their cost: previous slide.

Code, protocol, results and these slides: github.com/NinadHora/MM845 (willmore-audit/). The protocol document fixes every threshold and its justification.



Paper: [arXiv:2604.04321](https://arxiv.org/abs/2604.04321)



WillmorePINN code



MM845 course material



This project:
github.com/NinadHora/MM845

Appendix: validation on exact surfaces; grid resolution

Surface	W obtained (expected)	int K dA (expected)	Willmore residual
Sphere	12.5663706143 (4 π)	12.5663706143 (4 π)	7.6e-7, critical
Clifford torus	19.7392088023 (2 π^2)	4.6e-11 (0)	1.6e-6, critical
Torus R = 3	31.4048888987 (closed form)	2.7e-10 (0)	0.81, not critical: a valid torus that is not a Willmore torus

Grid resolution. On a surface with a thin feature, Gauss-Bonnet fails visibly on a coarse grid while W is still moving by tens of units; its reference value is exact and W's is not. So the harness refines every grid until Gauss-Bonnet closes to 1e-6, and reports the grid with each certificate.

On the two checkpoints: χ -hat = 2.000000 and 0.000000 (residuals 1e-14, 1e-15); Mobius with two centres: 1e-16; both below 8 π , so embedded by Li-Yau.

Appendix: limitations, and what was left out

Estimates are working days on a laptop CPU unless the cluster is named. The plan allowed stopping before every item below.

Item	Status and limitation	Estimate
Genus 2	not certified: the two-chart quadrature (disc removed from each chart, collar counted once) is not written; everything said about genus 2 is at the level of the reported energy	1 to 2 days
Basin map	9 initial data instead of 30 to 50; with 9 points the clustering says little, so Figure 2 is not shown	1 night on the cluster (slurm/basins.sbatch) + half a day
Attack ladder	rung 1 only (random search over bubbles); no bubble passes the strict acceptance rule (loss within 2 sd of the base), so no formal success; greedy is written, not run; annealing over the weights not written	greedy half a day; annealing 1 to 2 days
Published flow	used as released, not reimplemented (deliberate: the audit must be independent of the training loop, and a reimplementation adds no certificate)	not planned
C4 on a network	fourth derivatives of a tanh network are noisy; the threshold is a band, not a line; the variational fallback only sees smooth directions	inherent
Residual-loss PINN	weak form only (18 test directions); the pointwise form does not run in this torch build; diverged at lr 1e-3, stable at 1e-4 with clipping	pointwise form via jacrev: 1 day
Mesh test	96 x 96 grid; detects a 2% overlap, clean at a 5% gap; a near-touch below the grid scale is missed	finer grids: hours
C6 Kuwert-Schatzle	ran on one run only (run_1: 58 increases in 200 epochs); the other 16 runs have per-epoch checkpoints and take 30 seconds each	hours
C7 Weiner, Hessian index	not run: Hessian-vector products, Lanczos, Mobius and reparametrisation modes projected out; the gauge projection is the hard part	3 to 5 days
C8 Bobenko discrete energy	not run: the mesh now exists, the formula is short	half a day
Other arms	L-BFGS and Sobolev flows (half a day each); periodic CNN, set transformer, mesh flow (2 to 3 days each)	see left